

GiftKart: An Intelligent Multi-Vendor Gifting & E-Commerce Platform

Full-Stack Web Development | MERN Architecture (MongoDB, Express.js, React.js, Node.js)

Project Title:	GiftKart — Scalable Multi-Vendor Gifting Store & Admin CMS	Academic Year:	2025 – 2026 (Final Year Major Project)
Domain / Track:	Full-Stack Web Technologies & Distributed Systems	Degree / Branch:	B.Tech / B.E. / BCA / MCA (Computer Science / IT)
Primary Tech Stack:	MERN Stack, Tailwind CSS, REST APIs, JWT, bcrypt	Candidate / Guide:	Student: Vikash Kumar Guide: Project Supervisor

1. ABSTRACT

GiftKart is a modern, production-style, multi-vendor e-commerce web application engineered to bridge the gap between conventional commodity online marketplaces and specialized occasion-driven gifting. While standard e-commerce giants prioritize volume and generic logistics, they overlook critical nuances of gifting: bespoke laser engraving, custom handwritten calligraphy cards, luxury packaging (velvet boxes, floral wraps, wooden crates), target delivery date alignment for milestones, and dedicated recipient-based discovery. GiftKart resolves these challenges through a high-performance MERN (MongoDB, Express.js, React.js, Node.js) architecture. The system incorporates three synchronized portals: an interactive **Customer Storefront & Gifting Studio**, a dedicated **Multi-Vendor Seller Operations Hub** for merchant listing and warehouse dispatch, and an **Admin CMS** for fleet logistics, catalog curation, and delivery milestone orchestration. The platform features 100% Indian localization with standard Rupee (₹ INR) pricing, pincode SLA delivery estimation across Indian states, realistic simulated multi-channel payments (UPI QR, RuPay, Net Banking, COD), and printable GST-compliant tax invoices.

2. PROBLEM STATEMENT & MOTIVATION

Modern online consumers face notable limitations when attempting to purchase and deliver commemorative gifts on generic retail platforms:

- 1. Absence of Occasion-Specific Filtering:** Existing marketplaces treat gifts as standard retail SKU items, forcing users to manually sift through hundreds of irrelevant product listings without intuitive filters for Birthdays, Anniversaries, Weddings, or Festivals.
- 2. Missing Emotional Personalization:** Mainstream platforms fail to provide real-time interactive previews for custom laser engravings, uploaded photo plaques, or handwritten greeting notes prior to checkout.
- 3. Disconnect Between Sellers and Gifting Logistics:** Merchants lack unified dispatch workflows that generate consolidated gift-fulfillment packing slips and synchronized carrier AWB tracking numbers for time-sensitive surprise deliveries.
- 4. High Complexity & Third-Party Lock-In:** Many specialized boutique gifting platforms rely on expensive proprietary SaaS solutions, making them inaccessible for local artisans and Indian sellers seeking transparent marketplace onboarding.

3. PROJECT OBJECTIVES

The core technical and functional objectives of the GiftKart project are:

- **Architect a Scalable MERN Web Platform:** Develop a robust, decoupled 3-tier architecture with React.js frontend, Express/Node.js RESTful API backend, and MongoDB document persistence.
- **Design a Dedicated Gifting Studio:** Construct an exclusive /gifts hub featuring a 3-step Smart Gift Finder Wizard (Occasion → Recipient → Budget in ₹ INR) that delivers instantaneous client-side and server-side matching.
- **Implement Role-Based Access Control (RBAC):** Provide secure JWT-based authentication for three distinct user roles: Buyers, Verified Sellers, and Platform Administrators with protected API routes.
- **Empower Multi-Vendor Marketplace Operations:** Enable merchants to onboard via an Indian GSTIN-validated portal,

manage product catalogs (selling prices, MRPs, specifications, stock levels), and print warehouse fulfillment packing slips.

- **Simulate Real-World Indian Logistics & Payments:** Deliver an Ekart/BlueDart/Delhivery tracking simulator with public AWB lookup (/track), postal pincode delivery estimation, dynamic UPI QR scan-to-pay, card authorization, and GST invoices.

4. COMPARATIVE ANALYSIS: EXISTING VS. PROPOSED SYSTEM

Feature / Capability	Existing Generic Marketplaces	GiftKart Proposed Solution
Gifting Focus	Generic retail catalog; gifts mixed with general groceries/appliances.	Dedicated Gifting Studio (/gifts) exclusively for milestone celebrations.
Occasion Discovery	Manual keyword searches; no guided recommendation engine.	Interactive 3-Step Smart Finder (Birthday, Anniversary, Wedding, Festive).
Gift Customization	Very rare; requires contacting third-party sellers off-platform.	Integrated engraving text, photo upload preview, handwritten card notes.
Packaging Selection	Standard brown corrugated cardboard box.	Luxury Velvet Box, Floral Wrap & Wax Seal, Handcrafted Wooden Crate.
Seller Dispatch	Opaque enterprise portals with heavy licensing fees.	Streamlined Seller Hub with 1-click warehouse packing slips and AWB assignment.
Currency & Logistics	Fragmented currency switchers and third-party tracking links.	Standard Indian Rupee (₹ INR), Pincode checker, Ekart/BlueDart AWB milestones.

5. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

GiftKart is implemented as a 3-tier decoupled architecture separating presentation, business logic, and persistence layers. Communication between layers is conducted strictly via stateless JSON RESTful APIs over HTTP/HTTPS.

Architecture Layer	Technology / Library	Role & Academic Justification
Frontend (Client)	React.js (v19) + Vite + Tailwind CSS	Component-based reactive UI, client-side routing via React Router DOM, dynamic utility-first responsive layout styling.
State & Networking	React Context API + Axios	Centralized auth and cart management across components without heavy external state libraries; intercepted REST requests.
Backend (Server)	Node.js + Express.js	Asynchronous, non-blocking I/O event-driven web server hosting secure REST controllers, routing middleware, and data validators.
Authentication	JSON Web Tokens (JWT) + bcrypt.js	Stateless identity verification with cryptographic signature; salt-hashed secure password storage meeting OWASP guidelines.
Database (Storage)	MongoDB + Mongoose ODM	NoSQL schema-flexible document store optimizing rich nested product options, occasion arrays, order milestones, and seller data.

6. CORE FUNCTIONAL MODULES

- 1. User Identity & Role-Based Security:** Provides self-service registration and login. Supports role privileges (customer, seller, admin) using authorization middleware that checks JWT payload headers.
- 2. Dedicated Gifting Studio & Smart Finder (/gifts):** An exclusive discovery engine featuring Birthday Specials, Anniversary Keepsakes, Wedding Gifts, and Diwali Hampers. Features a 3-step interactive wizard filtering by Occasion, Recipient, and Budget in ₹ INR.
- 3. Multi-Vendor Seller Operations Hub (/seller/*):** Allows Indian sellers to register with a 15-character GSTIN, manage live catalog inventory, configure selling prices vs MRPs, track order dispatch queues, and print Flipkart-style warehouse packing slips.
- 4. Customization & Luxury Packaging Engine:** Allows buyers to configure personalized laser engraving text, recipient

names, photo plaque previews, handwritten greeting card notes, and gift wrapping options (Kraft box, Floral Wrap, Royal Velvet, Wooden Crate).

5. Indian Logistics & Fast-Track AWB Tracking (/track): Integrated 6-digit Indian PIN code delivery SLA estimator (Bangalore, Mumbai, Delhi, Chennai, Kolkata), live transit timeline simulator for Ekart Logistics, BlueDart, Delhivery, and DTDC.

6. Multi-Channel Indian Payment Gateway: Simulated multi-method checkout supporting instant UPI QR scan & pay (GPay, PhonePe, Paytm), RuPay/Visa cards with 3DS OTP verification, Indian Net Banking (SBI, HDFC, ICICI, Axis), and Cash on Delivery (COD).

7. Admin CMS & Operations Portal (/admin/*): Full management of system categories, catalog products, buyer orders, delivery fleet assignments, status transitions, and GST-compliant printable tax invoices.

7. DATABASE SCHEMA DESIGN

Entity / Collection	Key Attributes & Schema Types	Relationships & Constraints
User	name, email, passwordHash, role, sellerProfile: {storeName, gstin, phone, address}	Primary identity record; referenced by Product (seller) and Order (customer).
Product	name, category, price, mrp, stock, images, occasion, recipient, isGift, isAssured, seller	Belongs to Category and User (seller). Indexed on name and description for text searches.
Order	user, items[], customization: {engraving, giftBox, greetingCard}, totalAmount, payment, tracking	Belongs to User (customer); embeds shipping address, carrier AWB, and milestone timeline.
Category	name, slug, description, image, isActive	Referenced by Product; drives marketplace and gifting catalog taxonomy.

8. HARDWARE & SOFTWARE REQUIREMENTS

Category	Minimum Requirement	Recommended Environment
Operating System	Windows 10 / Ubuntu 20.04 / macOS 12	Windows 11 / Ubuntu 22.04 LTS (64-bit)
Processor	Dual-Core 2.0 GHz Intel / AMD	Quad-Core Intel i5 / AMD Ryzen 5 or higher
RAM	4 GB RAM	8 GB – 16 GB DDR4/DDR5
Storage	10 GB Free Storage Space	SSD with 25 GB+ Free Space
Runtime & Server	Node.js v18.x, npm v9.x	Node.js v20+ LTS, npm v10+
Database Server	MongoDB Community Server v6.0+	MongoDB Community v7.0+ / Atlas Cloud
Web Browser	Chrome / Edge / Firefox (HTML5 compatible)	Latest Google Chrome / Microsoft Edge

9. TESTING & VERIFICATION METHODOLOGY

1. Unit & Component Verification: Frontend React modules and UI subcomponents were verified for clean rendering and zero compiler warnings using Vite production builds (npm run build, 1697 modules transformed with zero errors).

2. RESTful API Integration Testing: Developed comprehensive automated integration test suites validating user registration, seller GSTIN onboarding, product creation, catalog retrieval, cart persistence, order creation in ₹ INR, and public AWB tracking lookups.

3. Security & RBAC Validation: Enforced and tested JWT authentication filters preventing unauthorized access to protected seller dashboard routes and administrative endpoints. Password hashes were verified using 10 bcrypt salt rounds.

4. Data Integrity & Currency Validation: Verified strict Indian Rupee (₹ INR) localization, mathematical accuracy of MRP strike-through percentage discounts, packaging cost additions, and GST tax invoice breakdowns.

10. FUTURE ENHANCEMENTS

- **Automated Milestone Reminders:** Recurring calendar notifications for user birthdays and anniversaries via web push notifications.
- **Augmented Reality (AR) Gift Preview:** In-browser WebXR preview allowing customers to project custom engraved items and bouquets on their physical table before placing orders.
- **Multi-Lingual Regional Support:** Expanding UI internationalization into Hindi, Tamil, Telugu, and Bengali to broaden accessibility across Tier-2 and Tier-3 Indian cities.

11. CONCLUSION

The **GiftKart** project successfully demonstrates the design and engineering of a full-stack, enterprise-grade e-commerce application tailored specifically to the multi-vendor Indian gifting ecosystem. By combining a dedicated celebration Gifting Studio, a Flipkart-style merchant operations portal, realistic simulated Indian logistics and payment workflows, and strict Rupee localization, GiftKart surpasses standard academic demonstration projects and serves as an industry-ready final year engineering capstone. All code is structured following modern modular patterns, documented, and published to the Git repository for evaluation.

Candidate's Signature:

Project Guide's Signature:

Head of Department (CSE/IT):

Date: _____

Date: _____

Date: _____